

Sai Sriya Vemuri

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Summary

Software Development Analyst with 2+ years of experience building backend services, ETL pipelines, RESTful APIs, microservices, and reporting workflows for financial data systems. Experienced across Java, Spring Boot, Python, SQL, MySQL, MongoDB, Snowflake, Spark, Jenkins, and Tableau. Recently completed a Master of Quantitative Management in Business Analytics at Duke University's Fuqua School of Business, with hands-on experience in machine learning, recommendation systems, data validation, dashboarding, and stakeholder-facing analytics.

Education

Duke University, The Fuqua School of Business <i>Master of Science in Quantitative Management: Business Analytics</i>	Durham, NC 2025 - 2026
SSN College of Engineering <i>Bachelor of Technology in Information Technology</i>	Chennai, India 2019 - 2023

Experience

Keysight Technologies Business Analyst Consultant – Capstone Project	Mar 2026 – Apr 2026
- Built a Python and SQL recommendation pipeline that transformed customer, product, and transaction data into ranked Next Best Offer outputs for cross-sell and upsell use cases. - Engineered and validated 55 reusable customer, product, and interaction-level features, improving ranking model inputs and enabling scalable recommendation experimentation.	
Citi Technical Analyst.	Jul 2024 – Jun 2025
- Built and maintained RESTful APIs and Java Spring Boot service-layer logic for financial reporting workflows, improving access to business-critical transaction data. - Improved backend data retrieval performance by 40% through MongoDB and MySQL indexing, query optimization, and validation improvements for high-volume reporting workflows. - Created reusable Python transformation and validation modules across 10+ ETL pipelines, reducing maintenance effort by 25%. - Operationalized Hadoop and Spark-generated outputs into production microservices, reducing reporting latency to under 10 minutes for business-critical transaction monitoring. - Implemented ISO 20022 CAMT message processing in Java-based services, increasing message coverage to 95% and improving downstream liquidity reporting consistency. - Developed a Java Excel export feature for 50K+ row downloads using streaming and pagination, maintaining p95 performance under 5 seconds. - Built standardized JUnit, Mockito, and golden dataset validation checks across API, service, and data layers, increasing automated test coverage by 30%. - Led a Snowflake proof of concept over 100K+ records, creating analytics-ready datasets with SQL validation and data quality checks for dashboard reporting. - Reduced failed releases by 20% by standardizing Jenkins CI/CD workflows for data pipeline and backend service deployments.	
Citi Automation Intern	May 2022 – Jul 2022
Created an internal AWS S3 automation tool for object uploads and retrieval, reducing manual processing effort by 75%.	

Skills

- Languages:** Java, Python, SQL, C++, JavaScript
- Backend:** Spring Boot, REST APIs, Microservices, Maven, Gradle
- Data and Databases:** MongoDB, MySQL, PostgreSQL, Hadoop, Apache Spark, ETL, Data Warehousing, Data Pipelines
- Analytics & BI:** Tableau, Power BI, Excel, Pandas, Scikit-learn
- DevOps and Deployment:** Docker, Jenkins, CI/CD, Bitbucket, Red Hat Linux, Shell Scripting
- Cloud:** AWS, Azure, GCP, Amazon EC2, S3
- Testing and Monitoring:** JUnit, Mockito, Kibana, Copilot, Unit Testing, Integration Testing
- AI/LLM:** RAG, FAISS, Prompt Engineering, FastAPI, Streamlit, LLM Application Development

Projects

- Jobstream: Backend Component for Distributed Processing** (Java, Spring Boot, PostgreSQL, Redis, MinIO):
 - Built Redis-backed async job processing with **LPUSH/BRPOP**, 5 job APIs, and support for 3 worker instances to scale background execution.
 - Managed a 6-service Docker Compose stack using Java and Spring Boot, adding retries, idempotency, and 7-state job tracking for reliable processing.
- BlogWeb: Full Stack Application** (Node.js, React, Express, MongoDB, Google OAuth):
 - Designed and implemented 13 Express routes and integrated **React** front-end components for authentication and blog workflows, using Mongoose ORM for end-to-end CRUD operations.
 - Strengthened application security with authenticated/guest middleware and ownership-based edit and update checks.
- Lead Scoring & Marketing Funnel Analytics** (MySQL, Spring Boot):
 - Designed dashboard-ready REST API over 84k+ Salesforce leads to compute funnel conversion, velocity KPIs, and segmentation.
 - Performed rule-based lead scoring to predict PQL, launch, and first shipment outcomes within 30/60/90-day windows.